

Bachelor Thesis

Automatically calculating up-to-date CO₂ emissions
from the combustion of liquid and gaseous fossil fuels
in Austria

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Elijah STÄNGL

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Supervisor

Assoc. Prof. Priv.Doiz.DI Dr.nat.techn. Johannes Schmidt
Institute of Sustainable Economic Development
Department of Economics and Social Sciences
University of Natural Resources and Life Sciences, Vienna

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Kurzzusammenfassung/Abstract

Um das selbstgesetzte Ziel von Klimaneutralität bis 2040 zu erreichen, müssen Österreichs Treibhausgasemissionen jedes Jahr deutlich sinken. Der größte Anteil an Emissionen stammt aus der Verbrennung fossiler Brennstoffe im Energiesektor. Offizielle Emissionsdaten werden mit einer bis zu zweijährigen Zeitverzögerung veröffentlicht, wodurch die Analyse aktueller Emissionstrends verzögert wird. Diese Bachelorarbeit präsentiert eine Methode, um aktuelle CO₂-Emissionen aus der Verbrennung von flüssigen und gasförmigen Brennstoffen bereitzustellen, und implementiert die Darstellung der daraus resultierenden Daten auf einer Onlineplattform. Die Emissionen werden aus regelmäßig aktualisierten Daten zur Brennstoffverbrennung berechnet. Mehrere Verarbeitungsschritte sind notwendig, um fehlende Verbrauchsdaten einzubeziehen und den nicht energetischen Verbrauch von fossilen Brennstoffen zu exkludieren. Die Ergebnisse unterscheiden sich in allen Kategorien um durchschnittlich deutlich weniger als 2% von offiziellen Emissionsdaten. Die Herangehensweise eignet sich gut, um langfristige und kurzfristige Emissionstrends mit geringer Zeitverzögerung zu analysieren. Zukünftige Arbeiten könnten Emissionen des Industriesektors einbeziehen.

Schlagworte: CO₂-Emissionen, Brennstoffverbrennung, Visualisierung von Emissionen, Emissionstrends

To meet the self-defined target of climate neutrality by 2040, Austria's greenhouse gas emissions need to decrease significantly each year. The largest share of emissions is fossil fuel combustion in the energy sector. Official emission data for Austria is published with a delay of up to 2 years, which postpones the analysis of emission trends. This bachelor thesis presents a method to provide up-to-date CO₂ emissions from the combustion of liquid and gaseous fuels and shows the resulting data on an online platform. The emissions are calculated with regularly updated data for fuel combustion. Various processing steps are necessary to address missing consumption data and exclude non-energetic uses of fossil fuels. The results differ from official emission data by substantially less than 2% on average across all categories. The approach works well to analyse emission trends with a short time lag. Future work may include emissions from the industry sectors.

Keywords: CO₂ emissions, fuel combustion, visualising emissions, emission trends

1 Introduction

Human-caused climate change already massively affects weather and climate extremes and impacts food and water security, human health and economies around the world [1]. Changes in Austria's climate have been accelerating, especially since 1980, impacting mean temperature as well as weather extremes. The observed temperature change in Austria is significantly greater than the global average [2]. Global warming mainly originates from the emission of human-caused greenhouse gases (GHG), with CO₂ and methane emissions dominating [1]. Austria's GHG emissions in 2024 were with 7.5 t CO₂ equivalents per person about 12% above the global average of 6.7 t/person [3]. Austria has specified the goal of achieving climate neutrality by 2040. To achieve this goal, a rapid change in energy infrastructure not yet seen in Austria's history would need to occur [4]. Austria's responsibility and the urgency of reducing GHG emissions are therefore particularly large.

In 2024, the energy sector accounts for 64.7% of all GHG emissions and 75% of all CO₂ emissions in Austria. Fuel combustion activities contribute to 99.1% of energy emissions. Most of these emissions are caused by CO₂, with 97.3% of total GHG emissions in the category energy fuel combustion being CO₂. GHG emissions in the energy sector are dominated by CO₂ much more than total emissions [5]. International aviation is not included in a country's total emissions, even though fuel combustion is occurring [6].

Official and extensive emission data for Austria is published by the Umweltbundesamt. The National Inventory Documents follow the IPPC guidelines and fulfil Austria's mandatory reporting towards the UNFCCC [7]. The document is released jointly with the Common Reporting Tables, which contain background data [8]. The latest instalment, Austria's National Inventory Document 2026 (NID), includes data until 2024 [5]. Emissions for the year 2025 have been forecasted by the Umweltbundesamt [9]. Nevertheless, there is a relevant gap between the occurrence of emissions and public data availability in Austria throughout the year.

The goal of this bachelor thesis is therefore to at least partly close this gap by providing a good approximation for current CO₂ emissions from the energy sector. Since liquid and gaseous fuel combustion contributes in 2024 together more than 93% of CO₂ emissions from fuel combustion in the energy sector [5], these are the two fuel types focused on. The goal is to make projections available on an online platform with data as up to date as possible. This leads to the following two research questions:

What is a possible technical solution to calculate and publish up-to-date and frequent CO₂ emissions from gaseous and liquid fuel combustion in Austria?

and

How accurate and up-to-date are these published CO₂ emissions?

The two main boundaries for this project are the following: Only CO₂ emissions, and not all GHG emissions, are included. The emissions from the energy sector are limited to gaseous and liquid fuels. Further explanations about the chosen system can be found in a further chapter (2.1 Concept and Data Availability).

The CO₂ emissions are calculated by using publicly available consumption data for different fuels. This raw data must be modified to exclude non-energy use and to account for missing data. The emissions are calculated from consumption data by applying generic conversion and emission factors. To test the accuracy of the calculations, yearly emissions are compared

to Austria's Common Reporting Tables 2026 (CRT). The code for these calculations is written in Python. Matplotlib and Pandas are the two extensions primarily used.

In Materials & Methods, the concept and data availability for this project will be explained in detail before giving an overview of the complete process of calculating and visualising the results. The subsequent chapter shows results, including a validation of the approach. In the discussion section, possible improvements are presented and results are critically assessed, before everything is summarised in the conclusion.

2 Materials & Methods

The following chapter explains the goals of this bachelor thesis in more detail and clarifies the scale and limitations of the project. Furthermore, the process of calculating CO₂ emissions in three different categories and visualising them will be described.

2.1 Concept and Data Availability

The general goal for this project is to calculate up-to-date CO₂ emissions from fuel combustion activities in Austria. Since it is convenient to have a reliable source for comparison to validate the accuracy of the calculations, the included sources are chosen to mimic reported emissions from Austria's National Inventory Document 2026 (referred to as NID in this bachelor thesis). To validate the calculations, they are compared to Austria's Common Reporting Tables 2026 (CRT), which are consistent with the NID [5]. To best meet the goal of the project, only CO₂ emissions reported within the CRT category 1.A *Fuel Combustion Activities* are supposed to be included within the calculations. CRT category 1 *Energy* investigates emissions from energy activities. The category 1.A includes all emissions from fuel combustion activities, excluding emissions from biomass fuels. Emissions from international transport through ships or aircraft, summarised as international bunker fuels, are not included in a country's total emissions, but are reported separately [6]. This category also does not include emissions from iron and steel production, which are reported under the category *Industrial Processes and Product Use*. The subcategory *Iron and Steel Production* is responsible for 15% of GHG emissions in Austria, and all emissions from this category are accounted for under *Industrial Processes*, regardless of their use as fuels [5]. The category 1.A can be split into subcategories. The three sources responsible for the most GHG emissions from the subcategories reported within the NID are, in order of importance, *Transport* (1.A.3), *Manufacturing Industries* (1.A.2) and *Energy Industries* (1.A.1) [5].

Furthermore, the project's scale is limited by the fuel types included in the calculations. In the NID, the fuels are listed separately for the categories liquid, solid, gaseous and waste, with liquid and gaseous accounting for 93.29% of CO₂ emissions [5]. The calculations within this bachelor thesis are therefore limited to liquid and gaseous fuels. Another boundary regarding the results of this bachelor thesis is that only CO₂ emissions and not all GHG emissions are included. CO₂ emissions were responsible for 97.33% of all GHG emissions in CO₂ equivalents in the year 2024 in the energy sector [5].

Figure 1 illustrates the share of total GHG emissions covered by these boundaries in recent decades. CO₂ emissions from the combustion of oil products and natural gas clearly represent well over 50% of all GHG emissions in Austria over the whole period.

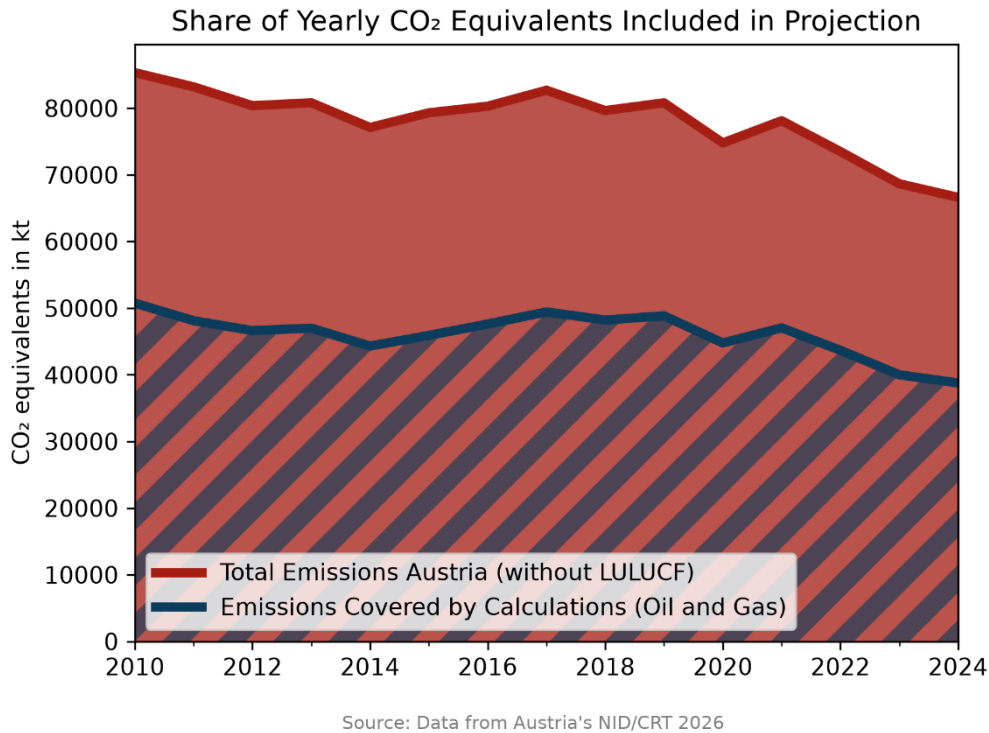


Figure 1: Share of total emissions covered

Although international aviation is responsible for about 80% of fuel consumption from aviation within the EU [10] and for 99.1% in Austria in 2024 [5], emissions from international aviation are often overlooked. One example of this is that emissions from international bunker fuels were not covered within the Kyoto Protocol [10]. While these emissions are not included in Austria's total emissions from fuel combustion [5], international aviation significantly contributes to global warming and is even predicted to expand [11]. Therefore, international aviation is also included in the calculations and reported separately.

Another main intention for this project is to automatically update the calculations for CO₂ emissions and make them publicly available. The results should therefore be automatically visualised and explained on an online platform by the end of this project [12]. To achieve all previously stated goals of this bachelor thesis, the data used for the calculations needs to be recent and regularly updated. Therefore, the publishing frequency of the underlying data must be higher than yearly.

2.2 Calculations for Different Fuel Categories

This chapter aims to transparently explain the calculations made to derive emission estimates. It reports on the data sources and how data is modified. For additional transparency, the created code is publicly available on GitHub [13].

The general method used for calculating CO₂ emissions from fuel combustion in this bachelor thesis follows a very simplified bottom-up approach. This means that activity data is used and multiplied by emission factors [14]. Therefore, the ideal, but unavailable, data would consist of the total consumption of natural gas and different oil products, excluding any non-energy usage. Since this data is not publicly available, assumptions had to be made to come up with reasonable estimates. This limits the accuracy of the calculated CO₂ emissions but still allows for showing general trends.

The conversion factors and emission factors for the calculations are generally taken from the NID. It is therefore important to transform all values into the same units. As the NID uses consumption data from the national energy balance and converts those into net calorific values (NCV), the same convention is followed here [5].

2.2.1 Calculations for Natural Gas

The obvious choice for data regarding natural gas consumption seemed to be a daily updated data set published by AGGM and E-Control. These two datasets showed similar, but not the same, consumption values in testing. Further in detail assessment of the data, however, led to the realisation that relevant consumption is missing from those datasets. A comparison provided by E-Control shows that a dataset provided by E-Control with the name “Inlandsgasverbrauch” matches the total consumption published by Austria’s energy balance and includes own consumption [15], which is probably mainly responsible for the missing gas consumption from the daily datasets, which potentially exclude own consumption. Data for gas consumption with a monthly update frequency, including the own gas consumption, are publicly available as a dataset and can be downloaded from E-Control [16]. Experience shows that the monthly data is missing the recent 1-2 months.

The generic formula used for the calculation is:

$$\text{CO}_2 \text{ emissions from natural gas} = (\text{inland gas consumption} - \text{non-energy use}) * \text{emission factor}$$

Since E-Control uses gross calorific values (GCV) for their data [17] and NCVs are needed for the following calculations, the gas consumption data is converted to NCV, using a conversion factor of 0.9 [15].

The second transformation needed, before CO₂ emissions can be calculated from the gas consumption data, is to subtract any natural gas used for non-energy purposes. The non-energy use of natural gas mentioned in the NID is feedstock use for ammonia production [5]. The natural gas used for ammonia production is reported in the NID for the years 1990 to 2024 [5]. Since the fluctuation within those values is relatively low, the non-energy use from natural gas consumption for this project is calculated by taking the arithmetic mean of all values since 2010, provided by the NID. This results in an annual gas use for ammonia production of 4.269 TWh. The monthly value is derived by dividing by 12, assuming that there is no seasonality within the ammonia production. This non-energy use is subtracted from the total gas consumption data provided by E-Control, as it does not count towards this CRT category [5].

In the NID, an emission factor of 55.6 t CO₂/TJ of natural gas is used for all years since 2019. Although the emission factor for the years before 2019 was slightly lower [5], the newest value is used for all calculations in this bachelor thesis, which simplifies the calculation. Transforming all values into the same units and multiplying the consumption of natural gas by the emission factor results in estimates of monthly CO₂ emissions.

The formula stated before, with applied values and unit transformations, looks as follows:

$$\begin{aligned} & \text{CO}_2 \text{ emissions from natural gas in kt} \\ &= \left((\text{inland gas consumption in TWh (GCV)} * 0.9) - \frac{4.269}{12} \right) * \left(3600 * \frac{55.6}{1000} \right) \end{aligned}$$

2.2.2 Calculations for Liquid Fuels

Since consumption data for petroleum products is not published by E-Control, another data source is required. Eurostat provides a lot of regularly updated consumption data for petroleum products for all EU countries, including Austria. The monthly data set, which seemed to best fit the objectives of this project, is the Observed Gross Inland Deliveries (GID) [18]. The definition for these values goes as follows:

“The observed delivery of finished petroleum products from primary sources (e.g. refineries, blending plants, etc.) to the inland market.” [19]

The definition of Calculated Gross Inland Deliveries defines the calculation as follows:

“Primary product receipts + Gross refinery output + Recycled products – Refinery fuel + Imports – Exports – International marine bunkers + Interproduct transfers – Products transferred + Stock changes” [19]

Since observed and calculated GID only deviate by a statistical difference [19], this calculation gives further information about the included products in the used dataset. *Stock changes* are defined as positive when stock is drawn [19]. This is the preferred form for calculating emissions, since they occur at the time of combustion.

There are two main problems with these definitions in comparison to an ideal data set. Firstly, some consumption is missing. Especially the unaccounted-for refinery fuel will lead to a gap in the consumption data, which cannot be closed easily without additional information. No suitable dataset was found to account for the missing emissions. Therefore, these emissions are incorporated using a simple approximation as a final step. It is important to note that a relevant gap between the uncorrected calculations and the NID is expected. The applied correction to account for these changes is presented below.

Secondly, non-energy use of petroleum products is not excluded from the raw data. Since the dataset includes a large variety of different fuel types [18], it is possible to exclusively use fuel types that are mainly used for fuel combustion. These are the fuel types included in the calculation for this project:

- gasoil and diesel oil (without biofuels)
- fuel oil
- liquefied petroleum gas (LPG)
- gasoline (without biofuels)
- kerosene (without biofuels)

The datasets without biofuels were used for the calculations because CO₂ emissions from biomass (including liquid biofuels) are generally not included in the emissions balance [5].

It is safe to assume that those fuel types are not completely combusted, while on the other hand not every fuel product excluded is completely used for non-energy consumption. The category *other oil products (without petroleum coke)* includes a variety of different products. Most of the products included in this category, according to the International Recommendation for Energy Statistics (IRES), like lubricants, bitumen and paraffin waxes [20], are distinctly mentioned as used for non-energy and feedstock in the NID [5]. Since a differentiation would be very difficult to implement, the decision was made to eliminate *other oil products* from this calculation.

Except for kerosene, all fuel consumption from the mentioned fuel types was used for calculating CO₂ emissions in this category. Kerosene is an exception, since it is primarily used

as jet fuel [20]. Emissions from flights can be separated into domestic and international aviation. Domestic aviation is included in CRT category 1.A and thus in this category. International aviation, however, is reported separately in the NID and not included in total emissions [5]. Kerosene consumption is accordingly split between the two categories. The data used for the NID clearly shows a trend in Austria towards an increasing share of aviation fuel consumption being used for international flights [21]. It would therefore not be beneficial to use an average share derived from all data points in the CRT to differentiate between international and domestic aviation, to calculate the split between domestic and international CO₂ emissions from aviation, as the kerosene used for domestic aviation is not a constant but trending [21]. Therefore, only recent data points are used, i.e. the mean of the last three data points reported in the CRT (2022, 2023, 2024). This results in a rounded value of 1.17% of kerosene consumption used for domestic aviation, which is included in this category and counts towards national Austrian emissions. The rest is accounted towards international aviation. This percentage is used for all years to simplify calculations.

A two-step process is necessary to calculate CO₂ emissions from GID with conversion and emission factors from the NID. The general formula used for calculating CO₂ emissions from a specific fuel type from the GID is the following:

$$\text{CO}_2 \text{ emissions from liquid fuels} = \text{gross inland delivery} * \text{conversion factor} * \text{emission factor}$$

The fuel consumption in the GID from Eurostat is reported in kilotonnes (kt). As a first step, this must be converted to joules with the individual conversion factor for this fuel type. For the calculations in this bachelor thesis, conversion factors from the annex of the NID are used [5]. The values used are country-specific conversion factors in TJ/kt for the year 2024. The values reported for 2024 are used for all years. The emission factors for liquid fuels are taken from the same table. The values used are default values for carbon emission factors in t C/TJ. Because emission factors for CO₂ are required, the values are multiplied by 44/12 [5] for the final calculations. The exact factors used for the calculations are shown in Table 1.

Table 1: Conversion and emission factors for liquid fuel combustion. Adapted from [5]

Fuel type	Conversion factor [TJ/kt]	Carbon emission factor [t C/TJ]
Gasoline	41.63	18.90
Jet Kerosene	43.40	19.50
Gas / Diesel Oil	42.50	20.20
Residual Fuel Oil	41.47	21.10
LPG	47.31	17.20

These calculations are performed for all five different fuel types. All emissions are afterwards summed up to obtain the total emissions from the combustion of liquid fuels. As expected, these CO₂ emissions are decidedly lower than the emissions reported in the CRT, since the data set used does not cover all relevant fuel consumption.

The missing emissions are not completely linear, as is visible in Figure 2. Nevertheless, as they do not show a clear trend, the average of the bias is used to correct total emissions in this category. The average yearly missing emissions in comparison to the CRT in the years 2013 to 2024 are 2862.42 kt CO₂, equivalent to 238.53 kt CO₂ per month. Since excluding these emissions from the estimates would result in relevant underreporting, this average

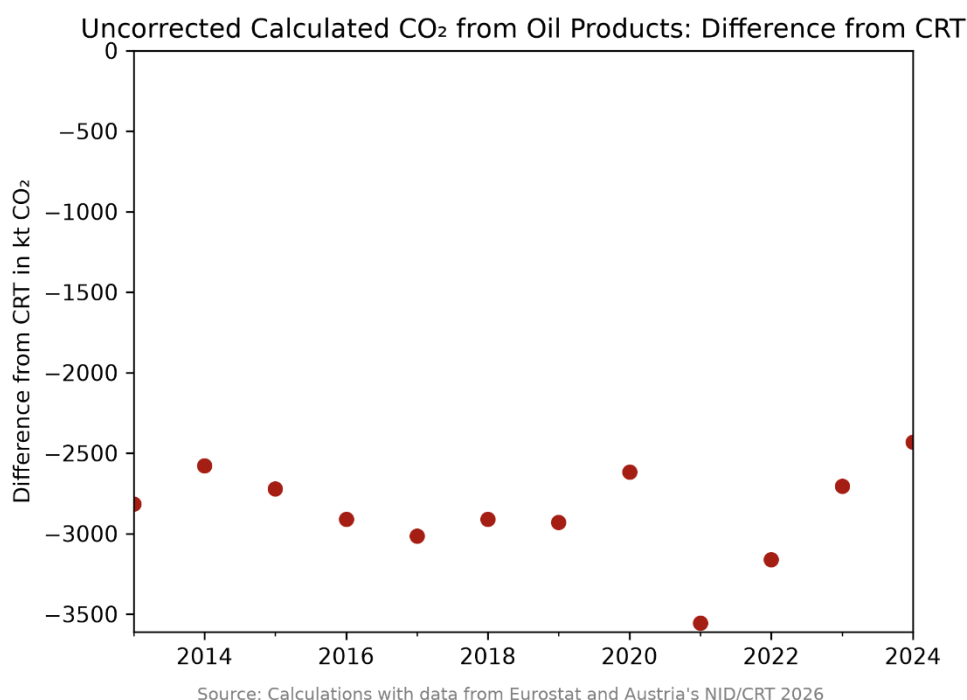


Figure 2: Missing emissions from liquid fuels

underestimation is added to the results. The monthly missing emissions of 238.54 kt CO₂ are rounded to 240 kt CO₂. These emissions are additionally added to the total emissions from liquid fuel combustion every month. The yearly emissions added are therefore 2880 kt CO₂.

2.2.3 Calculations for International Aviation

The calculation for international aviation is calculated from 98.83% of kerosene consumption (see previous chapter). The rest of the calculations follow the exact steps described for liquid fuels. The emission factors used for the calculation are also shown in Table 1 (kerosene).

2.3 Visualising Calculations for CO₂ Emissions

The final visualisations of the calculated CO₂ emissions mainly use the projections calculated as described in the previous chapters. For the comparison with the NID, monthly emissions are summed up to yearly values. To visualise the long-term trend of the emissions from different fuel types, an additional calculation is necessary. Seasonal variability limits the visibility of long-term trends, and thus, an additional 12-month rolling mean is calculated. This means that the value for every month consists of a mean of the last 12 months, including the one being reported. A 12-month rolling mean provides the advantage of showing emissions until the latest reported month but smoothens out seasonal variation.

In the comparisons with the NID, all years available in both datasets are shown. The graphs showing the seasonality of the current year and the three years before are presented. In all graphs showing long-term trends, the most recent 12 years are visualised. Those visualisations therefore show 144 values in every line.

3 Results

Here, the results of the estimations are presented in two main chapters: First, the projections are validated against data published in the CRT. Second, the visualisation of CO₂ emission projections, similar to how they are shown online, will be presented.

3.1 Comparison to Austria's National Inventory Document

Figure 3 illustrates the difference between the projections of yearly CO₂ emissions from natural gas and the values reported in the CRT for the gaseous category. The figure shows that the calculations result in slight overreporting in most years, with some outlier years (2022 and 2024).

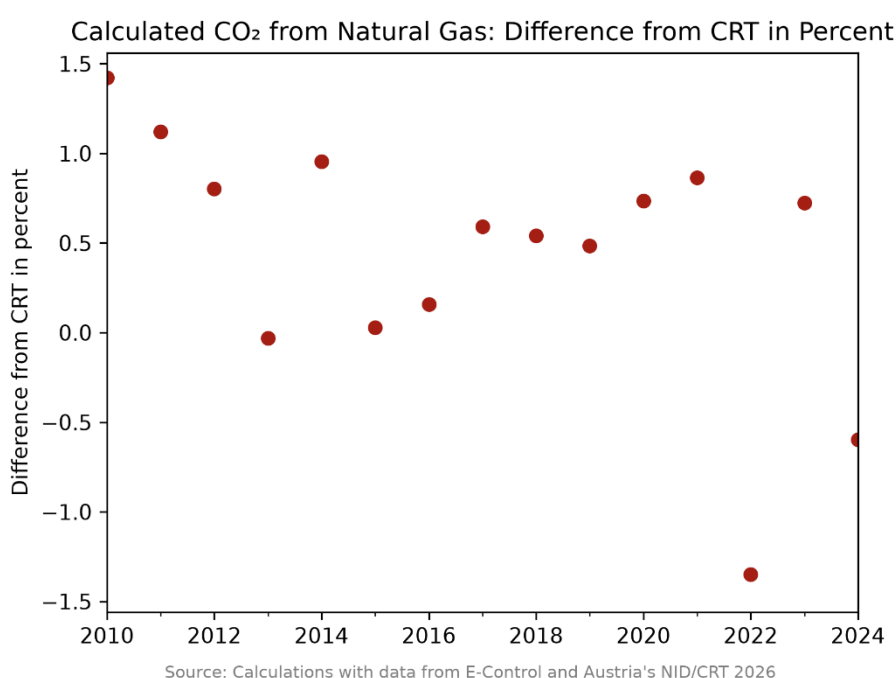


Figure 3: Difference from NID: Gas

The average absolute difference in the years 2010 to 2024 is 0.69%. The maximum difference is 1.42%. In comparison to the following two categories, the calculations for natural gas are closest to the NID.

For liquid fuel combustion, it is important to remember that the calculated values are already corrected with a constant to better match the NID. This was necessary because of missing consumption in the raw data (see chapter 2.2.2 Calculations for Liquid Fuels). Before this gap was corrected, there was an average underreporting of 9.55%.

Figure 4 illustrates the difference from the NID after applying the correction. There is no clear trend observable. However, the calculations match worse with the NID in more recent years. The average absolute difference is 0.75% and therefore similar to natural gas. The maximum difference of 2.25%, however, indicates a wider range of error.

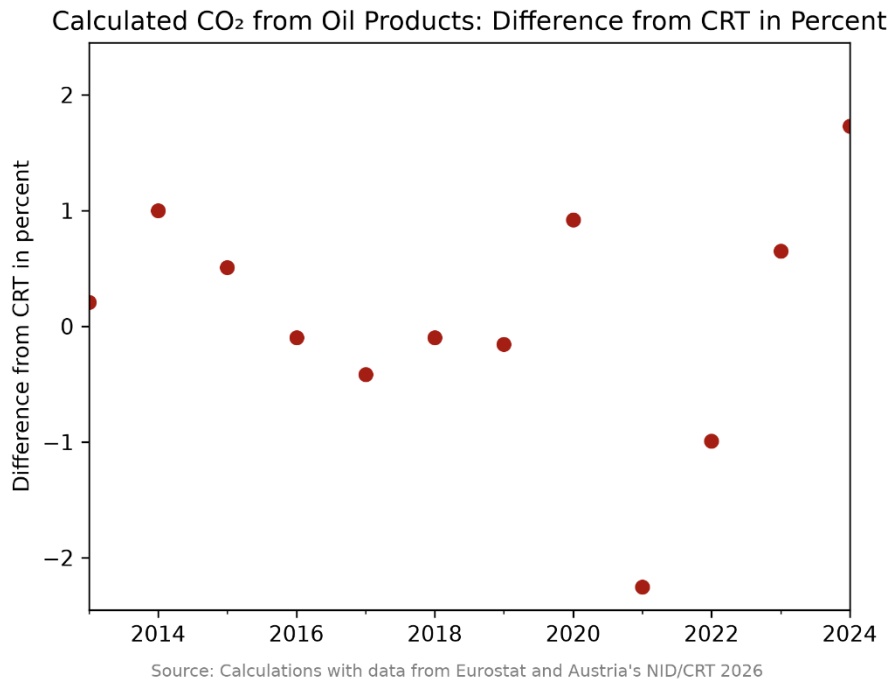


Figure 4: Difference from NID: Oil

Figure 5 illustrates the difference from the NID after combining CO₂ emissions from oil products and natural gas. The figure shows a smaller error compared to the individual comparisons. The average difference of 0.54% and the maximum difference of 1.12% clearly support this. This reveals the combined value of emissions from oil and gas as the most correct and reliable result from the approximations.

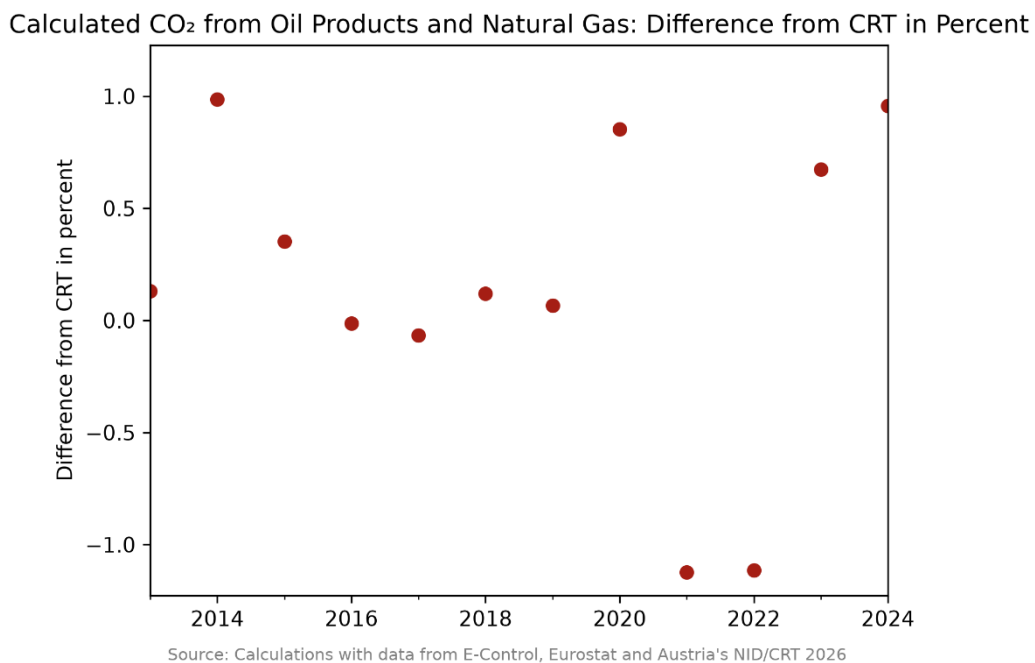


Figure 5: Difference from NID: Oil and Gas

International aviation is the least accurate category calculated in this bachelor thesis. This was expected because of methodological simplifications and the general difficulty of calculating international aviation emissions assigned to one country. Figure 6 shows a general trend of

overreporting in years longer ago towards underreporting in more recent years. This is explained by the trend towards a higher share of kerosene consumption used for international aviation and the constant adaptation factor used for the calculations (see chapter 2.2.2 Calculations for Liquid Fuels).

Calculated CO₂ from International Aviation: Difference from CRT in Percent

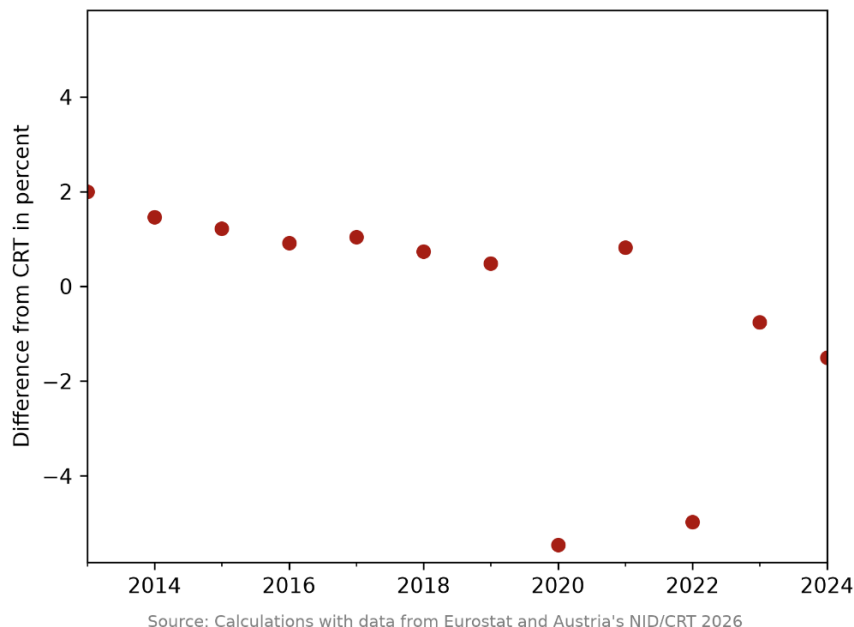


Figure 6: Difference from NID: International Aviation

There are two outliers visible in the years 2020 and 2022. Those values differentiate significantly from the NID. The average absolute difference is 1.78%, clearly above the two other categories. The maximum difference of 5.46% in the year 2020 shows a very large error range.

Figure 7 shows that although the calculations are the least accurate, the general trend of CO₂ emissions from international aviation is represented very well.

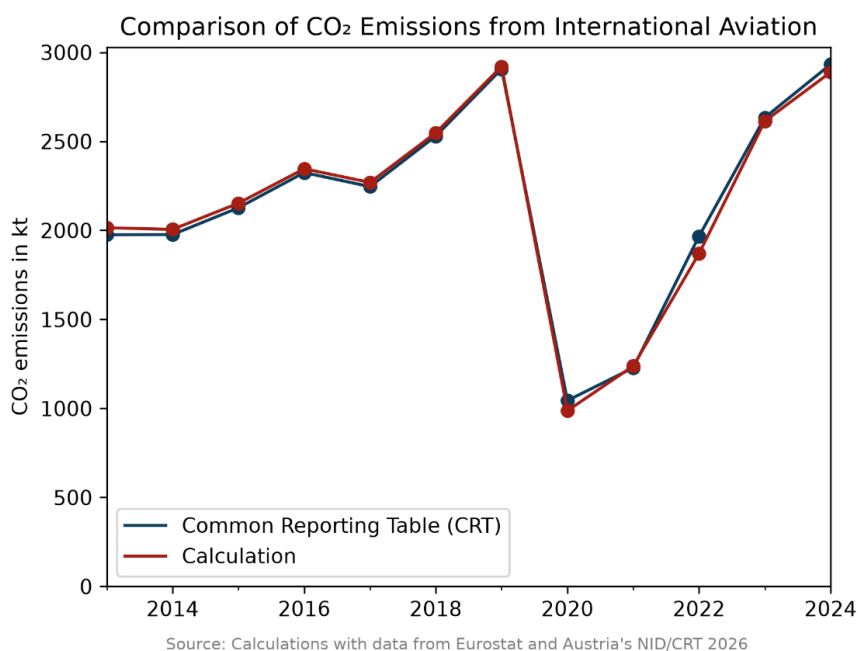


Figure 7: Comparison with NID: International Aviation

3.2 Visualising Monthly CO₂ Emissions

In the following chapters, the visualisations for the calculated CO₂ emissions are presented and interesting trends are highlighted. The analyses contribute to answering the research question in an indirect way by illustrating which conclusions can be drawn from the published figures. The visualisations published online always contain the latest published consumption data. Here, the figures are shown for illustration purposes. They contain the latest available data at the time of publishing. The latest month available is May 2026 for visualisations only containing natural gas and April 2026 for everything else.

3.2.1 Visualising Long-Term Trends Across All Categories

In Figure 8, the CO₂ emissions from all three calculated categories are visible. Every fuel category and combined emissions from oil and gas are visualised with two different lines. The lighter line with more variability shows the exact calculated CO₂ emissions for every month. The darker line represents the average over rolling 12 months (see 2.3 Visualising Calculations for CO₂ Emissions).

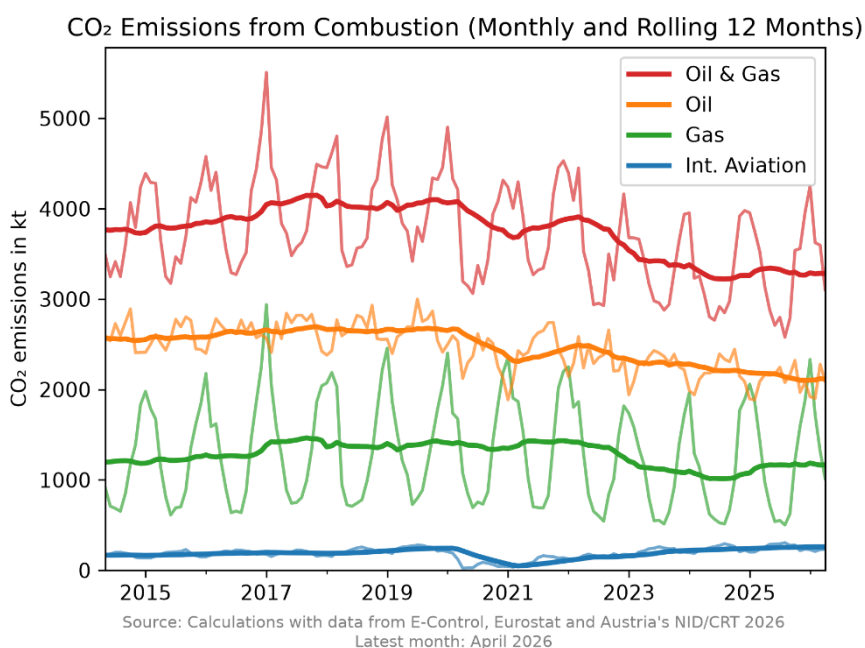


Figure 8: Long-term trend of all calculated CO₂ emissions together

The graph also indicates the amount of emissions between the three types of fuels. Oil is responsible for the most CO₂ emissions from combustion. Gas is also a major emitter with more than 1000 kt CO₂ per month. Emissions from international aviation are relatively low in comparison to the other categories. However, emissions from international aviation need more context and will be discussed later (4.2 Contextualising International Aviation). Furthermore, the general emission trends are clearly visible in Figure 8.

Figure 9 is better suited to analyse the emission trends of the three different categories separately. Both petroleum products and international aviation show a clear downward trend from 2020 to 2022. This is likely due to reduced energy consumption caused by the COVID-19 pandemic [22]. Especially international aviation shows a rapid downward trend, with emissions being down more than 75% over a short time span compared to the previous height.

CO₂ Emissions from Combustion (Monthly and Rolling 12 Months)

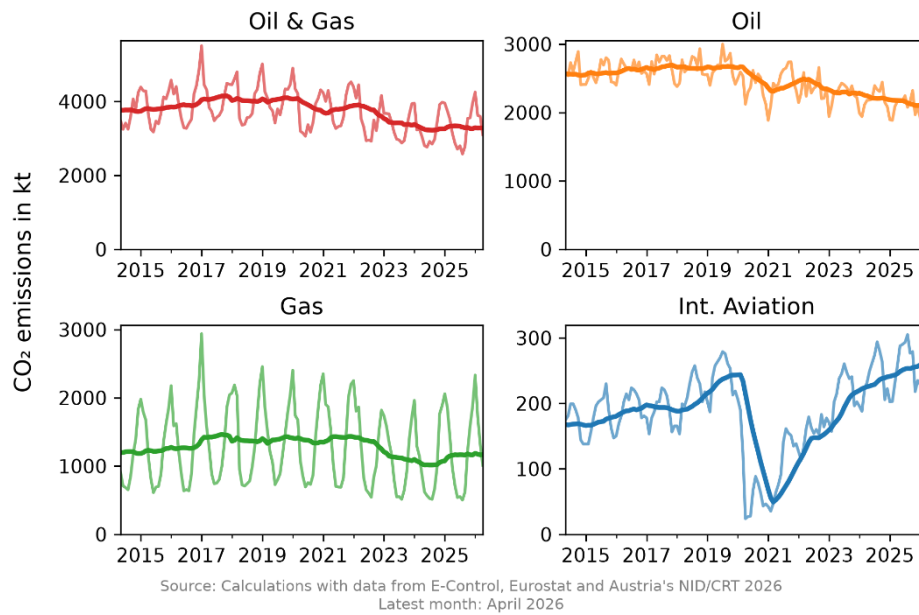


Figure 9: Long-term trend of all calculated CO₂ emissions separately

In recent years, however, emissions started growing again. In contrast, oil products show a clear downward trend after a short growth following COVID-19. Since oil products consist of several different fuels used for combustion, a more detailed analysis is presented in the next chapter.

CO₂ emissions from natural gas are more linear than the other two categories in the long-term trend. A change due to COVID-19 is not apparent. There is, however, a quite strong reduction in consumption in the years 2023 to 2025, clearly associated with the Russian invasion of Ukraine and the following price shocks [23]. The current average monthly emissions are slightly lower than those 12 years ago.

3.2.2 Visualising Long-Term Trends of Different Petroleum Products

To analyse the trends of different fuel types within the category of liquid fuels, they are separated into four different categories. Gasoil and diesel oil are grouped as one category, as they are also grouped in the raw data. Gasoline is shown separately. The category *Others* contains emissions from fuel oil, LPG and kerosene. The last category contains the missing oil consumption that was added with a constant. Therefore, no trend can be analysed here.

The area graph in Figure 10 is valuable for seeing the relevance of the different fuel categories for CO₂ emissions. Gasoil and diesel oil is clearly the category most relevant among all liquid fuels. Gasoline also accounts for a very significant share of all emissions. The category *Others* is not very important for an overview, especially in the more recent years. The missing oil combustion significantly alters total CO₂ emissions.

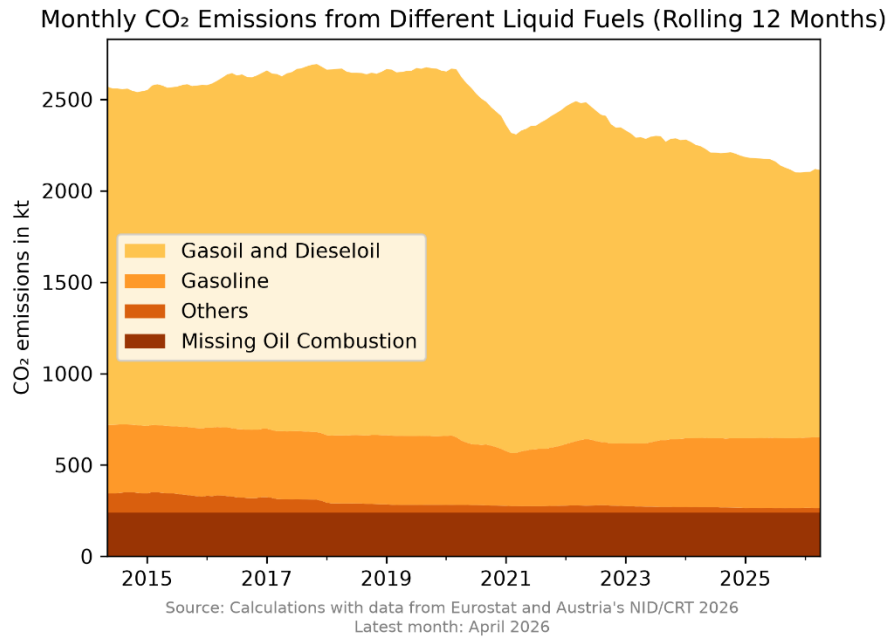


Figure 10: Long-term trend of different petroleum products: Area graph

Figure 11 is better suited to analyse the individual trends for the different fuels. The category *Others* follows a very steep downward trend since 2015. It is also evident that the high fluctuations in this category are only there until 2017. Looking into the data reveals that this trend is caused by the fade-out of fuel oil consumption. The category fuel oil refers to residual fuel oil and heavy fuel oil used in medium to large industrial plants and power stations [20].

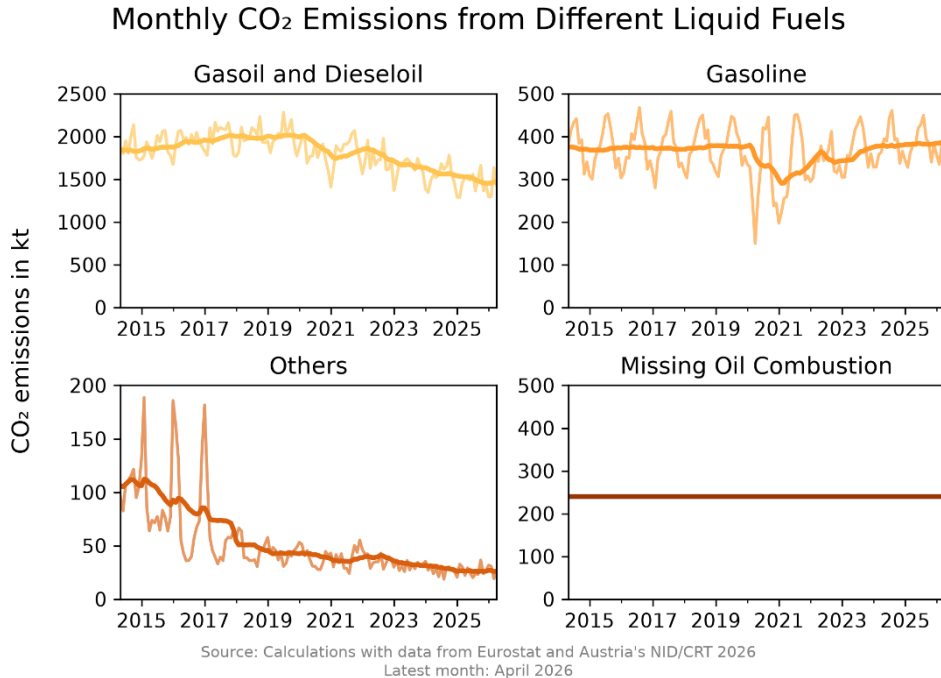


Figure 11: Long-term trend of different petroleum products: separate

Gasoline resembles a very stable trend in CO₂ emissions, with a clear decline during COVID-19. This matches a rather constant fleet of gasoline-powered vehicles in Austria over the last 20 years [24].

Emissions from gasoil and diesel oil follow a strong downward trend, which started during COVID-19. The trend towards lower CO₂ emissions seems to continue. Fuels in the category gasoil and diesel oil are primarily used for diesel engines and as heating oils. The long-term trend can be explained for both use cases. The fleet of diesel-powered vehicles in Austria stopped growing and started to decline slowly in the year 2018. The downward trend has become more rapid in recent years [24]. Data published by Statistik Austria [25] shows a more than 50% decrease in the installed base of oil heating systems over the last 20 years, with a constant period from 2013 to 2018. The combined decline in recent years explains the long-term trend seen in Figure 11.

In conclusion, the downward trend of CO₂ emissions from total liquid fuel combustion can be allocated towards falling emissions from the fuel types of gasoil and diesel oil. However, a relevant part of emissions is only estimated with a constant. Any trends from these missing fuel combustions cannot be observed from these calculations.

3.2.3 Visualising Recent Emissions

Another visualisation of the data is used to illustrate the trend in the current year more clearly. Another advantage of the chosen visualisation, which can be seen in Figure 12, is the visible seasonality.

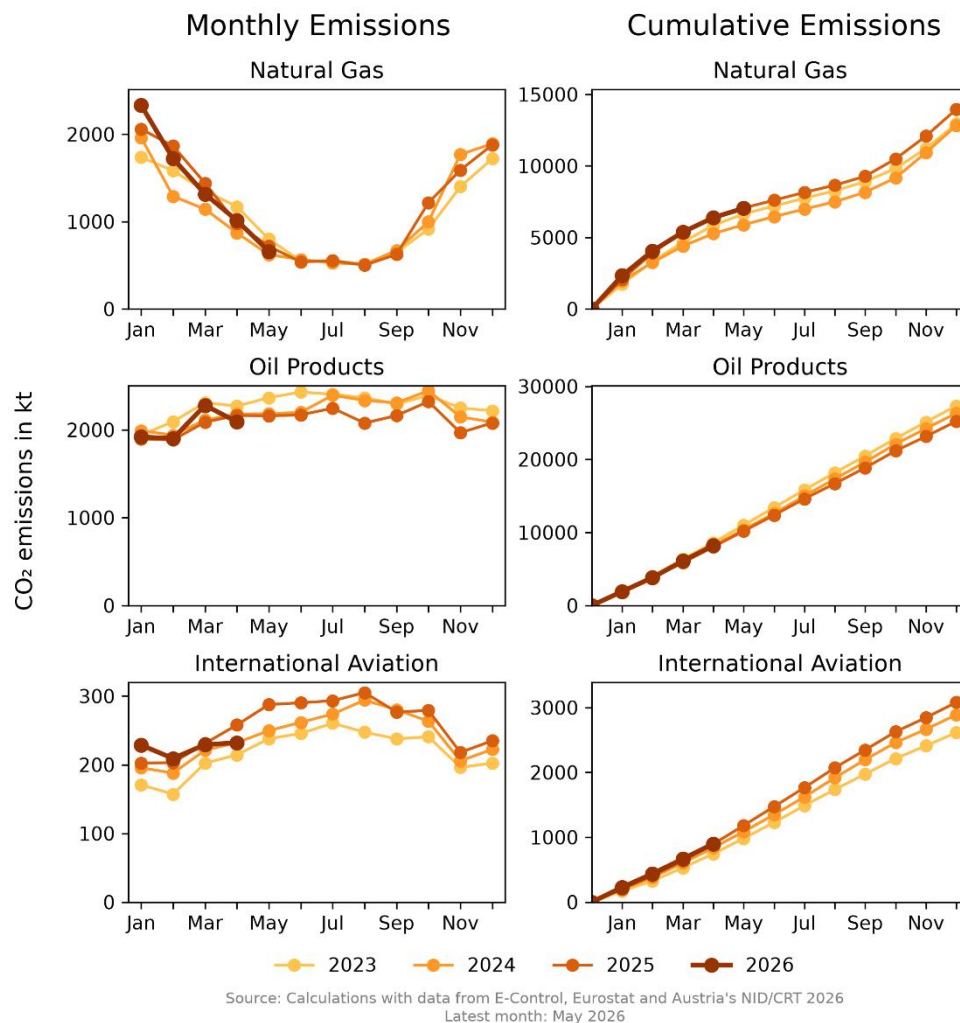


Figure 12: Current emissions from all categories

Figure 12 shows two different plots for every category. The monthly CO₂ emissions from the current year and the three years before can be seen on the left side. The right side shows the cumulative emissions for the same time span.

Natural gas has a very high seasonality because it is used for heating and for combined heat and power generation, which has a peak in winter. The current emissions are very similar to 2025 and higher than in the years before. Oil products show less seasonality throughout the year. The current emissions seem similar to those in 2025. International aviation shows slight seasonality with a big peak in the summer months and a small peak in December and January. This seasonality fits well with typical vacation times. The current CO₂ emissions from international aviation seem to continue the trend of rising emissions in recent years.

4 Discussion

This chapter gives context to the projections and proposes future improvements. First, the accuracy of the results will be discussed. Then, emissions from international aviation will be analysed in more detail. Afterwards, the completeness of the calculated emissions will be characterised.

4.1 Improving the Accuracy

All three categories allow learning about recent trends in CO₂ emissions from fuel combustion in Austria. The accuracy varies across the three categories. The average difference is well below 1% for oil and gas and under 2% for international aviation compared to official emission estimates.

The most important influence on accurate results comes from the raw consumption data used. If more accurate or better-suited data were available, this could lead to major improvements. The most important improvement could happen with consumption data for oil products, which contains all relevant oil consumption, so no data on missing consumption categories needs to be added as an estimate in the end. Although this was not achieved during this thesis, accounting for the missing fuel combustion may be possible by combining two datasets. Eurostat provides plenty more monthly datasets regarding oil consumption. One of those datasets directly addresses consumption in refineries. Directly adding this did not work out in simple attempts, but there may still be potential to experiment with this or other datasets provided by Eurostat. This could make the results more accurate and definitely more resistant in the future.

Another relevant factor is data processing. The conversion and emission factors used are currently directly taken from the NID. Since those factors are specifically for the year 2024, searching for adaptive or average factors could sharpen the results. If data were available that only contained fuels used for combustion, this would also eliminate major processing steps and reduce inaccuracies. The yearly consumption data from Eurostat provides much more detailed datasets. Some of those even directly differentiate between energy and non-energy use [26]. This could be used to better understand the usage of different oil products and possibly implement a more accurate differentiation.

The goal was to obtain a reasonable approximation of CO₂ emissions with as recent data as possible, and the approach delivers this. The deviation from official emission data is very low, but improvements are still possible.

4.2 Contextualising International Aviation

CO₂ emissions from aviation only show a very limited picture regarding climate impacts. Other important factors are nitrogen oxides, water vapour, aerosols and contrail formation. The climate warming potential through aviation is about three times higher when non-CO₂ impacts are included [27]. Although non-CO₂ emissions are relevant across all transport sectors and can also contribute towards climate cooling [28], it is important to mention this large difference regarding aviation to put the results into perspective. The temperature change per passenger-kilometer for air travel is much greater than for car travel on short to medium time scales. This

changes, however, when looking at longer horizons [29]. International aviation emissions should therefore not be overlooked, even though the category petroleum products, which is dominated by road transport [5], generates substantially higher CO₂ emissions. A very big share of emissions from air travel worldwide is accounted for by a very small percentage of the population [30]. This social inequality emphasises the importance of taking these emissions into account. Further work could compare emissions in terms of global warming potential. This would emphasise the entire climate impact and illustrate the importance of international aviation.

Another issue regarding international aviation is the question of how emissions should be allocated to specific countries. There are plenty of different options available. Some of those are: Allocation by fuel sold, by nationality of aircraft carrier, by the country of departure or arrival of the passenger, cargo or the aircraft. For some countries, the difference between the various options is very significant [10]. The NID uses a combination of these methods with adjustments to match the fuel sold reported in the national energy balance [5]. The method used for calculating international aviation in this project matches a simplified allocation by fuel sold but is in general in line with the NID.

4.3 Boundaries and Shortcomings

The fundamental goal for this project was to estimate CO₂ emissions from the energy sector in Austria. Due to limitations in data availability, some system boundaries had to be chosen. The emissions calculated only account for emissions from liquid and gaseous fuel combustion and closely resemble the CRT category 1.A. Furthermore, the calculated emissions are limited to the greenhouse gas CO₂. The chosen emissions, therefore, only include some part of all GHG emissions from the energy sector. For 2024, the emissions covered by the selected boundaries accounted for about 90% of total energy-sector GHG emissions in CO₂ equivalents, according to the NID. Since this sector accounted for 64.7% of total GHG emissions (excluding LULUCF) in the same year [5], the selected sources make up more than 58% of the total GHG emissions in Austria. This is only based on emission data for 2024, but it sets the calculations of this project into context. These percentages only refer to values in the categories liquid and gaseous fuel combustion. International aviation is not included there, but is added additionally for this project.

For future projects, it would be interesting to add more categories or greenhouse gases. A very relevant addition to represent a higher percentage of Austria's total GHG emissions would be the CRT category 2 *Industrial Processes and Product Use*. The iron and steel production alone accounted for 15% of Austria's total GHG emissions in 2024 [5]. Adding more emissions from this sector would definitely create interesting results. The data availability and realisability were, however, not examined in this thesis due to resource constraints.

The currently calculated data still has a small time lag of a few months. Future improvements could work on closing this gap even more. One possibility would be to explore if daily gas consumption data, which has significant flaws, could be used to project emissions from gas by means of a statistical learning model, linking monthly to daily data. Another possibility, which is possible with the currently implemented data, would be to create a projection for the last missing months. Very simple or complex methods could be used. A simple forecasting approach that extrapolates the current trend while accounting for seasonality could be implemented relatively easily and would allow for nowcasts of emissions.

5 Conclusion

Since GHG emissions in Austria are published with a time lag of more than a year, a more up-to-date approximation of current emission trends can provide interesting and useful information. This project succeeded in providing online available CO₂ emissions from liquid and gaseous fuel combustion in Austria. The research question for this bachelor thesis was with which technical solution this implementation is possible and how up-to-date and accurate the results are. The emissions are published monthly and the available data allows for calculations only missing the recent 1-3 months. Consumption data for natural gas is provided by E-Control and for petroleum products by Eurostat. The raw data is corrected for non-energy use and processed with conversion and emission factors from the NID. Because of incomplete consumption in the raw data, a constant value of 240 kt CO₂ per month is added to the category oil products.

The accuracy of the results varies depending on the category. The most accurate results were accomplished for natural gas combustion. Oil products provide a very good average, but include stronger variance in more recent years. International aviation is calculated with much lower accuracy, but still clearly demonstrates emission trends. The average difference from the official data is well below 2% in all categories. The results are visualised with multiple figures providing a good overview of long-term trends for the last 12 years and short-term trends for the current and recent years. The visualisations show a significant downward trend in emissions from oil products. This trend is induced by the fuel types gasoil and diesel oil. Emissions from gas consumption are much more linear, with no clear long-term trend. Emissions from international aviation still experience a strong increase, which started after a rapid decline during COVID-19. The presented results are well-suited to analyse long-term and short-term trends in Austria's fuel combustion emissions now and in the future.

Besides putting further effort into improving the accuracy of the results, a wider range of emissions could be included in further projects. There are several directions available to depict a wider range of GHG emissions. Because of Austria's relevant steel industry, including this would be especially interesting. Further additions could obtain other fuel types from the energy sector and emissions from other greenhouse gases than CO₂. The results from this project are a good start in providing more recent GHG emission data for Austria and in understandably visualising them. The potential for further additions and improvements is still abundant.

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Project Artefacts

The artefacts developed as part of this bachelor thesis are publicly available:

- Web Application: <https://elijahstaengl.github.io/co2-emissions-austria>
- Source Code: <https://github.com/ElijahStaengl/co2-emissions-austria>

The online platform always contains the latest approximations calculated.

The source code is provided in a GitHub repository.

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Use of Artificial Intelligence

Artificial intelligence was used while working on this project to improve the quality of the final result and to accelerate understanding and coding. The Document Analyzer provided by Boku University was used to find information in large reports. Grammarly helped with language correctness and clarity. ChatGPT was primarily used to help with coding techniques and sometimes to create small code blocks. Other use cases of ChatGPT were data and paper research, writing assistance and answering general questions.